

Recovering Stellar Classes from Photometric Structure in SDSS

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1 Introduction

The Sloan Digital Sky Survey (SDSS) provides a large sample of celestial objects classified as stars, galaxies, and quasars, together with photometric and spectroscopic measurements [1, 2]. In this project, the dataset serves as a useful setting for studying whether broad band observational features can recover meaningful astrophysical class structure. Because the known labels are defined spectroscopically while the main predictors in the analysis are photometric measurements, the central question is **whether these lower cost observational variables already contain enough information to distinguish among the three object types** [3]. This also makes SDSS a natural dataset for comparing unsupervised structure with supervised classification performance.

More specifically, this analysis is guided by four related questions:

- To what extent do the photometric measurements u, g, r, i, z exhibit a low dimensional structure that differentiates stars, galaxies, and quasars?
- Can unsupervised methods recover groupings that align with the known spectroscopic classes?
- How accurately can supervised classification methods predict the SDSS class labels from photometric features alone?
- How much does adding redshift improve class separability and classification performance?

To address these questions, the analysis uses exploratory data analysis and PCA to summarize the structure of the photometric variables, clustering methods to assess unsupervised class separation, and supervised classification models to evaluate how well the main SDSS features distinguish stars, galaxies, and quasars.

2 Data and preprocessing decisions

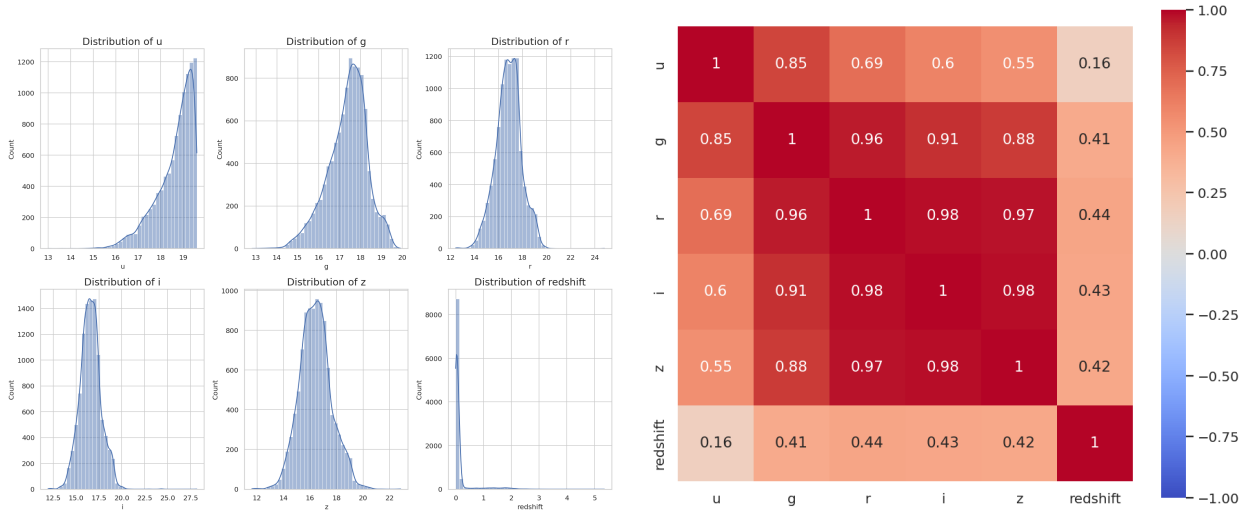
2.1 Data overview

SDSS is one of the largest astronomical surveys ever conducted, collecting imaging and spectroscopic measurements for millions of celestial objects. The dataset used in this project contains 10,000 observations from SDSS Data Release 14, where each object is labeled as a star, galaxy, or quasar based on spectroscopic analysis. Its main variables for multivariate analysis are the five photometric magnitudes u, g, r, i, z , which record brightness across wavelength bands from ultraviolet to infrared, together with spectroscopic redshift. The dataset contains no missing values, but the class distribution is imbalanced, with galaxies comprising 50.0% of the sample, stars 41.5%,

and quasars only 8.5%. The documentation also notes that the photometric variables and redshift are the most informative features for analysis, while several observation metadata variables are not directly useful for classification.

Figure 1 provides an initial overview of the main variables used in the analysis. As shown in panel (a), the five photometric magnitudes are concentrated within relatively narrow ranges, although their shapes are not identical: u is more strongly skewed, while g, r, i , and z are more tightly centered with a few extreme observations in the tails. By contrast, redshift has a markedly different distribution, with most observations clustered near zero and a long right tail extending to much larger values. This indicates that redshift captures a different aspect of the objects than the photometric variables and may provide additional discriminatory information in later stages of the analysis. The presence of skewness and a small number of extreme values also suggests that distributional shape should be kept in mind when interpreting downstream results. Appendix Figure A1 provides the summary statistics, which are consistent with the patterns in Figure 1.

Panel (b) shows that the photometric magnitudes are strongly positively correlated, especially among g, r, i , and z , indicating substantial redundancy in the original feature space. This makes dimension reduction a natural next step, since much of the variation in these variables is likely driven by a smaller number of shared directions. Redshift is only moderately correlated with the photometric bands, which suggests that it is not simply duplicating the same information. Taken together, these patterns motivate treating the photometric magnitudes as the core multivariate feature set, while considering redshift separately as an additional scientifically meaningful variable in later clustering and classification analyses.



(a) Marginal distributions of the five photometric magnitudes (u, g, r, i, z) and redshift.

(b) Correlation heatmap for the five photometric magnitudes and redshift.

Figure 1: Overview of the main SDSS features used in the analysis. Panel (a) shows that the photometric bands are relatively concentrated, whereas redshift is strongly right skewed. Panel (b) shows strong positive correlations among the photometric bands, especially among g, r, i , and z , but only moderate correlations between redshift and the photometric variables. Together, these patterns support using the photometric magnitudes as the core multivariate feature set, while treating redshift as an additional informative variable in later analyses.

2.2 Variable selection and preprocessing decisions

For the core multivariate analysis, the primary feature set consists of the five photometric magnitudes u, g, r, i , and z . These variables are the main observational measurements of object brightness across wavelength bands and form the natural basis for studying low dimensional structure in the dataset. Redshift is retained as an additional scientifically meaningful variable, but is treated separately from the main photometric feature set in order to assess whether it provides information beyond the broad band magnitude measurements.

The sky coordinates ra and dec were not used as primary modeling variables, since they describe object position on the sky rather than intrinsic physical properties relevant to class separation. Likewise, the observation metadata variables run , $camcol$, $field$, $plate$, mjd , and $fiberid$ were excluded from the core analysis because they primarily reflect the survey’s measurement and identification process rather than astrophysical structure. This choice keeps the analysis focused on variables with direct scientific relevance to distinguishing stars, galaxies, and quasars.

No missing value treatment was required because the dataset contains no missing observations. In addition, potentially extreme photometric values were retained in the initial analysis rather than removed a priori, since the dataset documentation notes that these are features of the SDSS pipeline output rather than preprocessing artifacts. Continuous variables were standardized before applying methods that depend on scale, such as PCA, clustering, and classification, so that differences in variance would not dominate the results.

These choices also shape the scope of the methods used later in the report. Because the main question concerns class recoverability from a single set of observational features, the analysis focuses on PCA, clustering, and supervised classification rather than methods designed for other structures. In particular, factor analysis was not used as a primary method because the objective is not to infer continuous latent traits, and canonical correlation analysis is not well suited here because the dataset does not provide a natural split into two variable blocks.

2.3 Multivariate normality assessment

Before applying PCA, the five photometric magnitudes (u, g, r, i, z) were assessed for approximate multivariate normality (See Figure A2 in Appendix). The results show very strong departures from a multivariate normal distribution. Mardia’s skewness test was highly significant ($\chi^2 = 5.05 \times 10^6$, $df = 35$, $p < 0.001$), and Mardia’s kurtosis test also strongly rejected multivariate normality ($z = 2.69 \times 10^4$, $p < 0.001$). These results indicate that the joint distribution of the photometric variables is far from Gaussian.

This conclusion is also supported by the Mahalanobis QQ plot, which shows substantial deviation from the 45 degree reference line, especially in the upper tail. While observations with smaller Mahalanobis distances remain closer to the theoretical χ^2_5 quantiles, the largest distances rise far above the reference pattern, indicating the presence of extreme observations and a much heavier tail than would be expected under multivariate normality. Taken together, the formal tests and the QQ plot provide consistent evidence that the photometric feature space is not well described by a single multivariate normal distribution.

This lack of multivariate normality is not unexpected in the present setting. The SDSS sample

combines stars, galaxies, and quasars within a common feature space, so a single homogeneous Gaussian population would not be expected a priori. In addition, the marginal distributions shown earlier already suggested skewness and tail behavior in several variables. For this reason, PCA is used in the next section primarily as a descriptive dimension reduction tool for summarizing the dominant correlation structure of the photometric variables, rather than as evidence for an underlying Gaussian generative model.

3 Principal Component Analysis of Photometric Structure

3.1 PCA setup

To examine the dominant structure of the photometric feature space, PCA was applied to the five photometric magnitudes (u, g, r, i, z) . The analysis in this section focuses on these variables alone, since they form the main correlated feature block in the dataset and are the natural starting point for dimension reduction. Before applying PCA, all five variables were standardized so that differences in scale and variance would not disproportionately influence the resulting components. Redshift was not included in the main PCA at this stage and is instead considered later as an additional variable in comparative analyses.

3.2 Component selection

Figure 2 summarizes the variance explained by each principal component. The scree pattern is dominated by the first component, which explains 87.5% of the total standardized variance, while the second component explains a further 11.5%. Together, the first two components account for approximately 99% of the total variation in the five photometric magnitudes, whereas the remaining three components contribute negligibly. This sharp drop after PC1, followed by the very small contribution of PC3 onward, indicates that the photometric feature space is effectively close to two dimensional for the purposes of interpretation.

Based on this pattern, the analysis that follows focuses primarily on the first two principal components. Retaining two components preserves essentially all of the variance in the photometric variables while still allowing the structure of the data to be visualized directly. Later components were not examined in detail because their explained variance is extremely small and therefore unlikely to contribute substantially to the main structural patterns of interest.

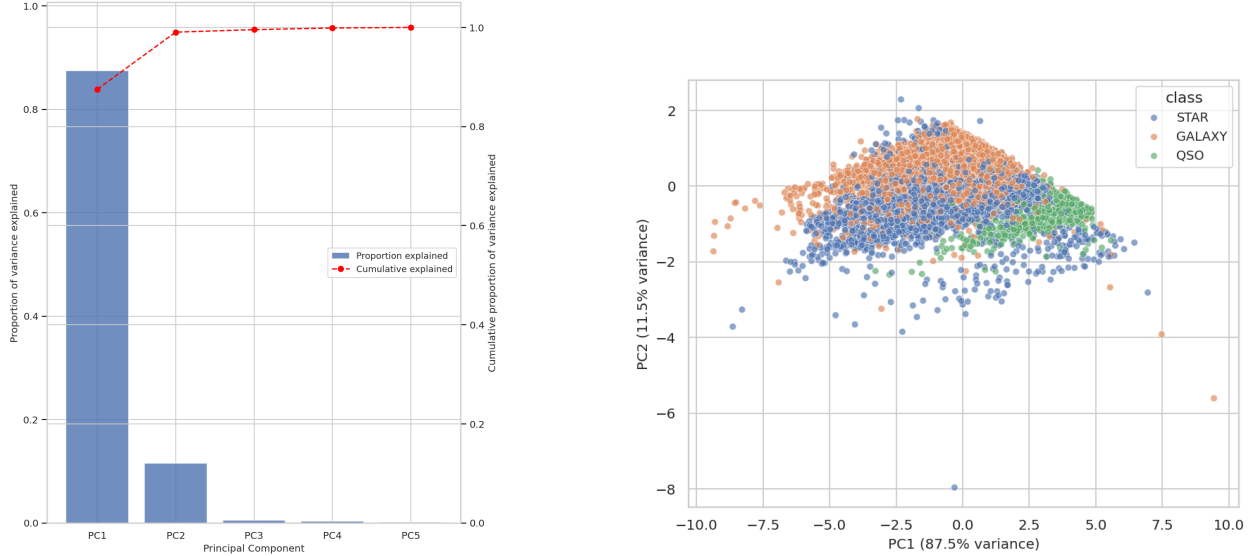
3.3 Interpretation of loadings and scores

Figure A3 in Appendix shows that PC1 reflects variation shared across all five photometric magnitudes. All variables load positively on this component, with similar values between 0.37 and 0.47, indicating that PC1 captures a broad common trend across the full set of bands rather than dependence on any single variable. Together with the variance results, this suggests that the dominant photometric structure is largely one dimensional.

PC2 captures the main secondary pattern after this shared trend has been removed. Its largest loading is on u at 0.82, while i and z load negatively, indicating that PC2 mainly reflects contrast between the ultraviolet band and the longer wavelength bands.

These patterns are also visible in the score plot in Figure 2(b). Quasars are more concentrated in a

relatively distinct region with larger PC1 scores, whereas galaxies and stars overlap more strongly in the center of the plot. Galaxies tend to lie higher along PC2, while stars are more diffuse and extend further downward. Overall, the first two components already reveal meaningful class structure, but the overlap between groups shows that photometric variation alone does not yield fully distinct class boundaries. This motivates the later clustering and classification analyses.



(a) Proportion and cumulative proportion of variance explained by the first five principal components.

(b) PC1–PC2 score plot for the photometric variables, colored by spectroscopic class.

Figure 2: PCA summary for the five standardized photometric magnitudes (u, g, r, i, z). Panel (a) shows that PC1 explains 87.5% of the total variance and PC2 explains 11.5%, so the first two components capture nearly all variation in the photometric feature set. Panel (b) plots observations in the first two PC score dimensions and shows partial class separation: QSOs cluster more distinctly toward the right side of the plot, while stars and galaxies overlap more strongly. Together, these results indicate that the photometric variables are highly redundant and largely low dimensional, while still retaining some class relevant structure in the leading principal components.

4 Unsupervised clustering and alignment with spectroscopic classes

4.1 Clustering setup

The PCA results show visible but incomplete class separation in the reduced photometric space, motivating a more formal unsupervised analysis. Clustering is performed on the first two principal component scores, which retain approximately 99% of the total variance in the five photometric magnitudes and preserve the main geometric structure visible in the photometric feature space. This low dimensional representation provides a natural basis for assessing whether the dominant photometric variation corresponds to meaningful group structure. Gaussian mixture models are used as the main method because they allow cluster specific covariance structure and probabilistic assignments, making them better suited than k means to the elongated and overlapping patterns seen in the PC score plot. K means is included as a simpler benchmark.

As shown in Appendix Figure A4, both AIC and BIC decrease sharply from $K = 1$ to $K = 2$, indicating that the PC score distribution is more structured than a single homogeneous cluster. Although both criteria continue to decline for larger K , the improvement becomes much more gradual, so they do not point to a clearly preferred small cluster number. In this setting, $K = 3$ is chosen not because it minimizes the information criteria, but because it provides the most direct comparison with the three known spectroscopic classes. The three component GMM is therefore used as an interpretable benchmark for class recovery rather than as the globally optimal density model.

4.2 Gaussian mixture models

Gaussian mixture models were fitted to the first two principal component scores to assess whether the dominant low dimensional photometric structure could be partitioned into clusters resembling the known spectroscopic classes. The fitted three component solution identifies overlapping elliptical groups that primarily divide the PCA space along the dominant PC1 direction. Figure 3(a) summarizes these results through the fitted clusters and covariance ellipses, the class composition within each cluster, and a direct comparison between the true spectroscopic labels and the GMM assignments in the PC1–PC2 plane.

The fitted GMM identifies three overlapping elliptical components that primarily divide the PCA space along the dominant PC1 direction. This indicates that the model is capturing broad density structure in the reduced photometric space rather than recovering three sharply separated groups. The comparison with the true spectroscopic labels (Figure 3(b)) confirms that the alignment is only partial. In particular, no single cluster corresponds cleanly to quasars, and the fitted clusters remain mixtures of stars and galaxies. The row normalized composition heatmap shows that clusters 0 and 1 are both dominated by galaxies but still contain substantial proportions of stars, while cluster 2 is dominated by stars but also contains a large fraction of galaxies. Quasars form only a modest share of cluster 0 and a very small share of the other two clusters.

Taken together, these results suggest that the first two principal components do contain meaningful structure, but that this structure does not map directly onto three well separated Gaussian subpopulations. Instead, the photometric PCA space appears more continuous and overlapping, especially for stars and galaxies, with quasars occupying a more localized region without forming an entirely distinct cluster under the GMM fit. This makes the GMM results useful descriptively, but not fully satisfactory as a recovery of the known SDSS class labels.

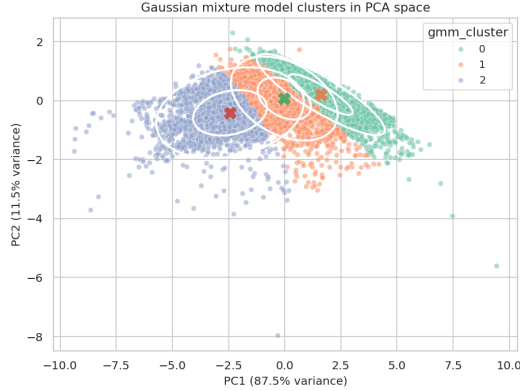
4.3 Comparison with k means

K means was applied to the same PC1–PC2 representation as a benchmark against the three component GMM solution. In Figure A5, the two methods produced broadly similar three part partitions of the reduced photometric space, but their boundaries were not identical. In particular, GMM represented the data using overlapping elliptical components, whereas k means partitioned the space into sharper regions based on Euclidean distance.

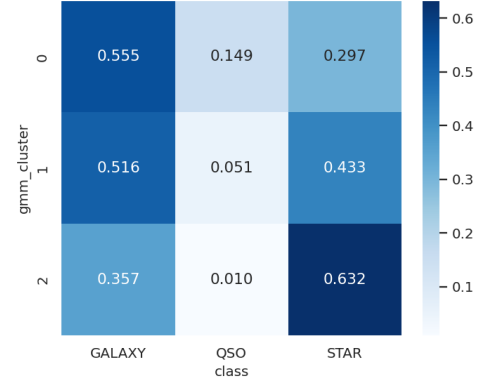
This difference is consistent with the objectives of the two methods. K means assigns each observation to the nearest centroid and therefore tends to divide a continuous feature space into compact regions. GMM instead fits a mixture of Gaussian densities with cluster specific covariance structure, so its goal is to capture the overall shape of the point cloud rather than produce the partition

that most closely resembles the observed class labels. In the present dataset, the PC1–PC2 space appears as a largely continuous and overlapping structure, especially in the region where stars and galaxies mix most strongly. For this reason, it is not surprising that the two methods yield related but noticeably different partitions.

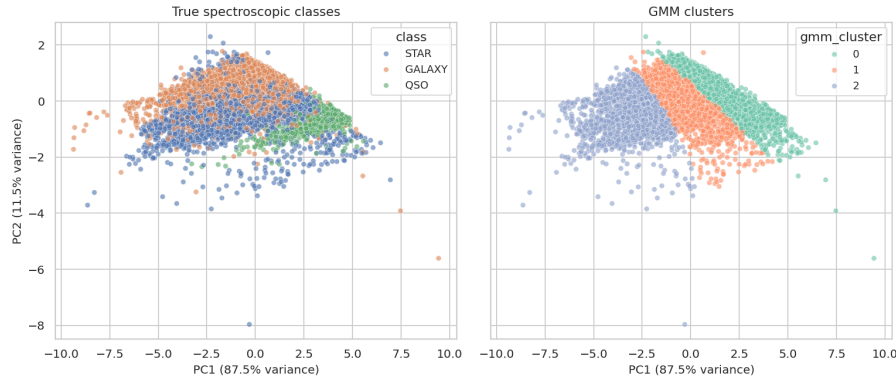
In Figure 3(c), both methods recover broad structure in the reduced photometric space, but neither provides a clean unsupervised reconstruction of the three spectroscopic classes. The comparison therefore suggests that the apparent grouping in PCA space is only partially stable across clustering methods and should be interpreted descriptively rather than as definitive class recovery.



(a) Gaussian mixture model clusters in the first two principal component dimensions, with cluster means and covariance ellipses overlaid.



(b) Row normalized composition of true spectroscopic classes within each GMM cluster.



(c) Comparison of the PC1–PC2 structure under true spectroscopic labels and GMM cluster assignments.

Figure 3: Gaussian mixture model results for the SDSS photometric data in the two dimensional PCA representation. Panel (a) shows the three component GMM fitted in the PC1–PC2 space, with ellipses indicating the estimated covariance structure of each mixture component. Panel (b) shows the row normalized class composition within each fitted cluster, indicating that all three clusters remain mixtures of stars and galaxies, while quasars are relatively rare in every cluster. Panel (c) compares the true spectroscopic labels with the fitted clustering in the same PCA space. Overall, the GMM captures broad geometric structure in the leading photometric PCA dimensions, but this structure is not closely aligned with the true object classes.

5 Supervised classification of SDSS classes

5.1 Classification setup

Because the unsupervised analyses showed only partial alignment between the photometric structure and the known spectroscopic classes, the next step is to assess how accurately these labels can be recovered under supervision. Two feature sets are considered: the five photometric magnitudes (u, g, r, i, z) alone, and the same variables augmented with redshift to test whether it provides additional discriminatory information beyond the photometric measurements alone.

For each feature set, the data were divided into training and testing subsets using a stratified train–test split so that class proportions were preserved across both sets. Continuous predictors were standardized before model fitting. Multinomial logistic regression was used as the primary supervised method because it provides a flexible discriminative baseline for multiclass prediction, while linear discriminant analysis was included as a comparison model with stronger distributional structure.

5.2 Multinomial logistic regression

Multinomial logistic regression was first fitted using the five photometric magnitudes (u, g, r, i, z) as predictors. As shown in Figure 4(a), the model achieved an overall accuracy of 0.934 on the test set, indicating that the photometric variables alone already contain substantial information about the spectroscopic class labels. Class specific recall was 0.957 for stars, 0.922 for galaxies, and 0.887 for quasars.

The confusion matrix shows that most observations were classified correctly, but the remaining errors were not evenly distributed across classes. Galaxies were most often confused with stars, with 7.0% of true galaxies assigned to the star class, while stars were misclassified as galaxies at a lower but still noticeable rate of 3.6%. Quasars also showed some overlap with the other two groups, with 6.1% predicted as stars and 5.2% predicted as galaxies. These patterns are consistent with the earlier PCA results, where quasars occupied a more localized region but stars and galaxies remained substantially mixed in the central part of the reduced photometric space.

Overall, the logistic regression results show that the five photometric magnitudes alone already support strong supervised recovery of the SDSS class labels, especially for stars and galaxies. At the same time, the weaker recall for quasars indicates that photometric information alone is still not sufficient for equally strong performance across all three classes. This suggests that the class signal is clearly present in the photometric variables, but it remains worth checking whether the same conclusion holds under a classifier with stronger distributional assumptions.

5.3 Comparison with LDA

To assess whether the supervised findings depend strongly on the choice of classifier, linear discriminant analysis was fitted using the same photometric feature set and the same training–testing split. As shown in Figure 4(a), LDA achieved an overall accuracy of 0.927, which is very close to the 0.934 obtained by multinomial logistic regression. Class specific recall was also similar across the two models: for LDA, recall was 0.941 for stars, 0.921 for galaxies, and 0.892 for quasars.

Although the overall difference between the two methods is small, the confusion matrices in Fig-

ure 4(b) show slight differences in error patterns. Relative to logistic regression, LDA performs marginally worse for stars but slightly better for quasars, while the two models are nearly identical for galaxies. The main point, however, is that both methods lead to the same qualitative conclusion: the photometric variables alone already contain substantial class information.

This agreement is notable because the two classifiers rely on different modeling perspectives. Multinomial logistic regression is a discriminative method that models class probabilities directly, whereas LDA is a generative linear classifier with stronger distributional assumptions. Although Section 2.3 showed strong departures from multivariate normality, LDA still achieves competitive accuracy, suggesting that the class signal in the photometric variables is strong enough to remain useful under this simpler model. At the same time, the assumption violations may help explain why LDA performs slightly worse overall than logistic regression. Taken together, these results suggest that the main supervised findings are not strongly method dependent. The next section examines whether this conclusion changes materially when redshift is added as an additional predictor.

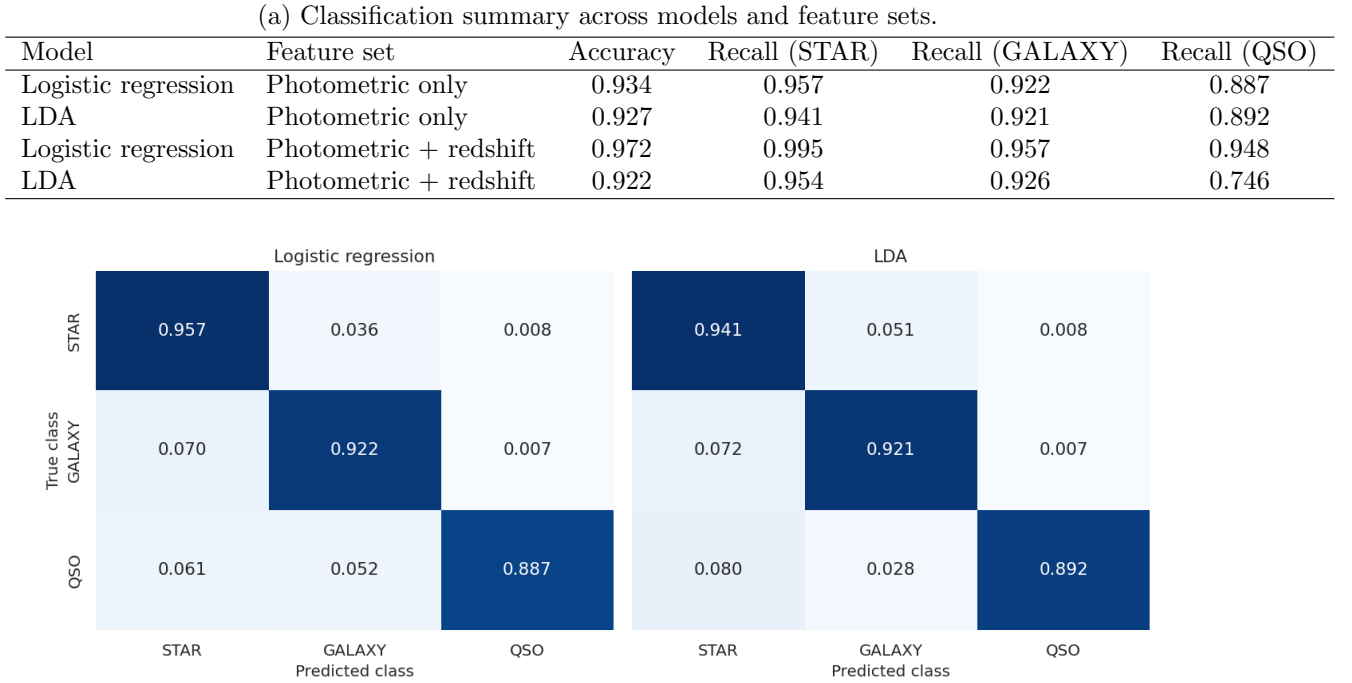


Figure 4: Supervised classification results for the SDSS spectroscopic classes using the photometric variables only. Panel (a) summarizes overall accuracy and class specific recall for multinomial logistic regression and LDA. Panel (b) shows the corresponding row normalized confusion matrices. Both models classify stars and galaxies well, while QSO classification is weaker. Logistic regression performs slightly better overall, especially for STAR recall, whereas LDA gives slightly higher recall for QSO. These results suggest that the photometric magnitudes alone support strong but not perfect class separation.

6 Robustness and sensitivity checks

6.1 Cross validation

To assess the stability of the supervised results, five fold stratified cross validation was applied to both multinomial logistic regression and LDA under the two feature sets. As shown in Figure 5(a), the main conclusions remain unchanged across folds. For logistic regression, mean accuracy was 0.933 ± 0.001 using the photometric variables alone and increased to 0.971 ± 0.002 after adding redshift, indicating that the gain from redshift is stable rather than split dependent.

The class specific recall results show the same pattern. Under logistic regression, adding redshift improved recall for all three classes, with the largest gain for quasars, from 0.871 to 0.941. By contrast, logistic regression and LDA were very similar under the photometric only feature set, but diverged after redshift was added, with LDA performing noticeably worse for quasars. Overall, the cross validated results show that the main supervised findings are stable across folds rather than dependent on a particular train–test split.

6.2 Sensitivity to feature set

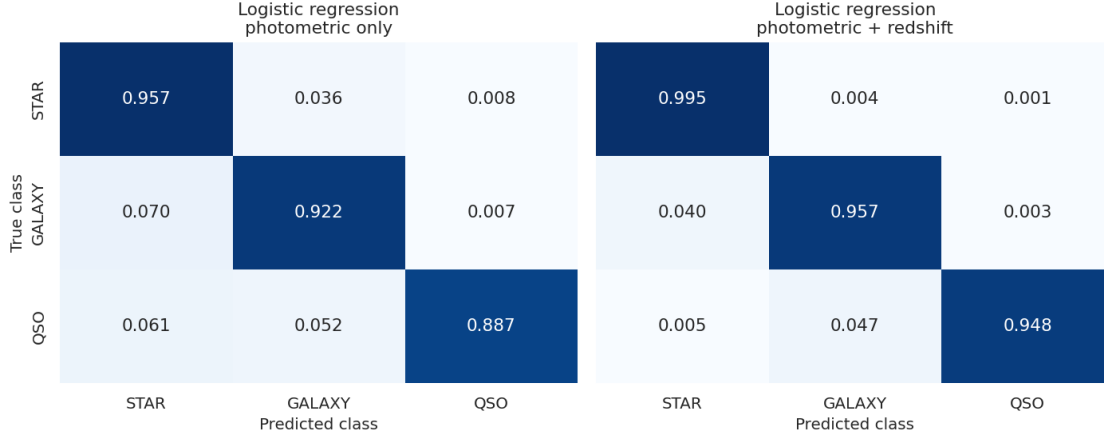
Figure 5(b) shows that the supervised results are highly sensitive to whether redshift is included in the predictor set. Under multinomial logistic regression with the photometric variables alone, the main errors arise from residual confusion among all three classes, particularly between stars and galaxies and, to a lesser extent, between quasars and the other two groups. After adding redshift, these off diagonal errors become much smaller across the confusion matrix.

The improvement is visible for all three classes. For stars, the misclassification rate into galaxies drops from 3.6% to 0.4%. For galaxies, the fraction misclassified as stars falls from 7.0% to 4.0%. For quasars, the fraction misclassified as stars decreases sharply from 6.1% to 0.5%, while misclassification as galaxies also declines slightly from 5.2% to 4.7%. These changes show that redshift does not merely fine tune the classifier, but adds complementary information that makes the class boundaries substantially cleaner, especially for stars and quasars.

This sensitivity to feature set is consistent with the earlier exploratory analysis. Redshift had a markedly different marginal distribution from the photometric variables and was only moderately correlated with them, so it was not simply duplicating information already present in the broad band magnitudes. Taken together, the supervised results suggest that the SDSS classes are already strongly predictable from photometric structure alone, but that redshift provides an additional and genuinely distinct source of discriminatory information.

(a) Cross validated classification performance across models and feature sets.

Model	Feature set	Accuracy	Recall (STAR)	Recall (GALAXY)	Recall (QSO)
Logistic regression	Photometric only	0.933 ± 0.001	0.954 ± 0.003	0.926 ± 0.006	0.871 ± 0.028
LDA	Photometric only	0.921 ± 0.003	0.929 ± 0.006	0.923 ± 0.006	0.871 ± 0.031
Logistic regression	Photometric + redshift	0.971 ± 0.002	0.994 ± 0.002	0.958 ± 0.006	0.941 ± 0.022
LDA	Photometric + redshift	0.921 ± 0.006	0.947 ± 0.007	0.927 ± 0.004	0.756 ± 0.049



(b) Row normalized confusion matrices for logistic regression with photometric variables only and with photometric variables plus redshift.

Figure 5: Robustness and feature set sensitivity of the supervised SDSS classification results. Panel (a) reports mean cross validated accuracy and class specific recall for multinomial logistic regression and LDA under the photometric only and photometric plus redshift feature sets, showing that the main supervised conclusions are stable across folds. Panel (b) compares the corresponding row normalized confusion matrices for multinomial logistic regression under the two feature sets. Adding redshift substantially reduces off diagonal errors across all three classes, with especially strong improvement for QSO classification. Together, these results show that the gain from redshift is both stable and substantively important, rather than a marginal improvement tied to a particular train–test split.

7 Conclusion

This project examined how well SDSS object classes can be recovered from the main observational features in the dataset, with a particular focus on the five photometric magnitudes and the additional contribution of redshift. The results show that the five photometric variables alone already contain substantial information about the SDSS class labels, although this information is expressed more clearly under supervision than under unsupervised structure recovery. In the PCA analysis, the first principal component explained 87.5% of the total variance and the second explained a further 11.5%, so the first two components together retained approximately 99% of the variation in the photometric feature space. This indicates that the photometric variables are highly structured and strongly redundant, with most of their variation concentrated in a very low dimensional representation.

At the same time, the unsupervised results show that this low dimensional structure does not translate into a clean recovery of the known spectroscopic labels. The PCA score plot revealed

visible class patterning, but with substantial overlap, especially between stars and galaxies. The GMM analysis identified broad density structure in the reduced space, but the fitted clusters did not correspond cleanly to the three observed classes. In particular, no single GMM cluster was dominated by quasars, and all three clusters remained mixtures of stars and galaxies to varying degrees. The comparison with k means further suggested that the apparent grouping in PCA space is not fully stable across clustering methods, rather than reflecting three naturally separated subpopulations. Taken together, these results imply that the photometric space is structured, but not in a way that supports clean unsupervised reconstruction of the true SDSS classes.

Supervised classification yielded a much stronger recovery of the labels. Logistic regression and LDA gave broadly similar results under the photometric only feature set, showing that the photometric variables alone already provide a strong basis for supervised prediction, even though they do not support clean unsupervised separation. In other words, the class information is present, but it is more effectively recovered when the labels are used directly.

The most important improvement came from adding redshift, which perfectly illustrates why comparing different methods is valuable. In the earlier PCA and clustering analyses, quasars were difficult to isolate because they overlap heavily with stars and galaxies in terms of broad band brightness. Redshift provided the crucial missing dimension. When redshift was included in the logistic regression model, overall accuracy increased from 0.934 to 0.972, and QSO recall saw a massive jump from 0.887 to 0.948. The confusion matrices showed that off diagonal errors shrank across all three classes. This is consistent with the exploratory analysis, where redshift showed a markedly different distribution. In the supervised setting, this unique information translated into substantially cleaner class separation.

The robustness checks supported the same overall conclusion. Five fold cross validation showed that the improvement from redshift was stable across folds rather than dependent on a particular split of the data, with the largest gain again occurring for quasars. Logistic regression and LDA remained broadly similar under the photometric only feature set, but diverged more clearly after redshift was added, suggesting that the benefit of redshift is both real and somewhat method dependent.

Overall, the analysis supports three main conclusions. First, the SDSS photometric variables already contain strong low dimensional and class relevant structure. Second, that structure is not naturally organized into three clean unsupervised groups, so clustering methods recover the known labels only partially. Third, supervised methods can recover the class labels much more effectively, and their performance improves substantially once redshift is included. The main limitations of the analysis are the overlap between stars and galaxies in photometric space, the relatively small proportion of quasars, and the clear departures from multivariate normality and from simple Gaussian cluster structure. Even so, the combined evidence shows that SDSS class recoverability is high under supervised methods, while redshift provides genuinely complementary information beyond the five photometric magnitudes and leads to substantially cleaner class separation.

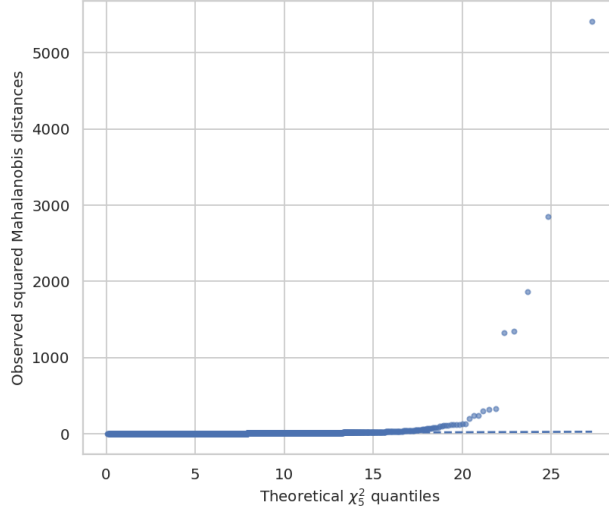
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Appendix

Variable	Mean	Std	Min	Median	Max
u	18.619355	0.828656	12.988970	18.853095	19.599900
g	17.371931	0.945457	12.799550	17.495135	19.918970
r	16.840963	1.067764	12.431600	16.858770	24.802040
i	16.583579	1.141805	11.947210	16.554985	28.179630
z	16.422833	1.203188	11.610410	16.389945	22.833060
Redshift	0.143726	0.388774	-0.004136	0.042591	5.353854

Figure A1: Summary statistics for the six continuous SDSS variables used in the analysis. The five photometric magnitudes (u, g, r, i, z) have broadly similar central tendencies and relatively concentrated ranges, whereas redshift is more skewed and spans a much wider range. This contrast suggests that redshift may contribute information that is not captured by the photometric variables alone.



(a) Mahalanobis QQ plot for the five standardized photometric magnitudes under a χ^2_5 reference distribution.

Statistic	Value
Mardia skewness	3028.881
Skewness chi-square	5048136.000
Skewness df	35.000
Skewness p-value	0.000
Mardia kurtosis	4541.587
Kurtosis z	26932.010
Kurtosis p-value	0.000

(b) Mardia skewness and kurtosis test results for the photometric variables.

Figure A2: Assessment of multivariate normality for the five standardized photometric magnitudes (u, g, r, i, z). Panel (a) compares observed squared Mahalanobis distances with the theoretical χ^2_5 quantiles expected under multivariate normality. The strong upward deviation in the upper tail indicates substantial departure from the reference distribution, driven by extreme observations and heavy tails. Panel (b) reports Mardia's skewness and kurtosis statistics, both of which yield effectively zero p-values. Together, these results provide strong evidence against the multivariate normality assumption for the photometric variables.

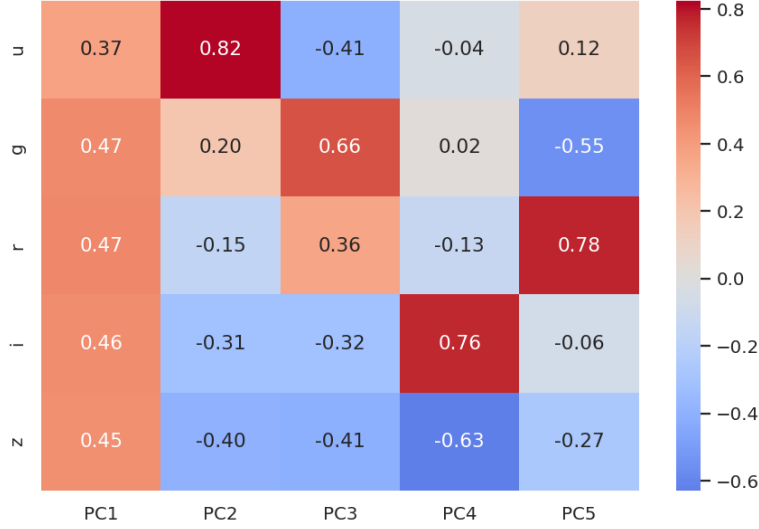


Figure A3: Heatmap of PCA loadings for the five photometric magnitudes (u, g, r, i, z) across the first five principal components, computed from the standardized photometric feature matrix. Each cell reports the loading of one variable on one component, with warmer colors indicating positive contributions and cooler colors indicating negative contributions. PC1 loads positively and fairly evenly on all five bands, indicating a broad shared brightness pattern across the photometric variables. In contrast, later components are driven by contrasts among bands, especially u versus g on PC2 and r versus z on PC5, showing that after the dominant common signal is removed, the remaining components capture wavelength specific differences rather than overall intensity.

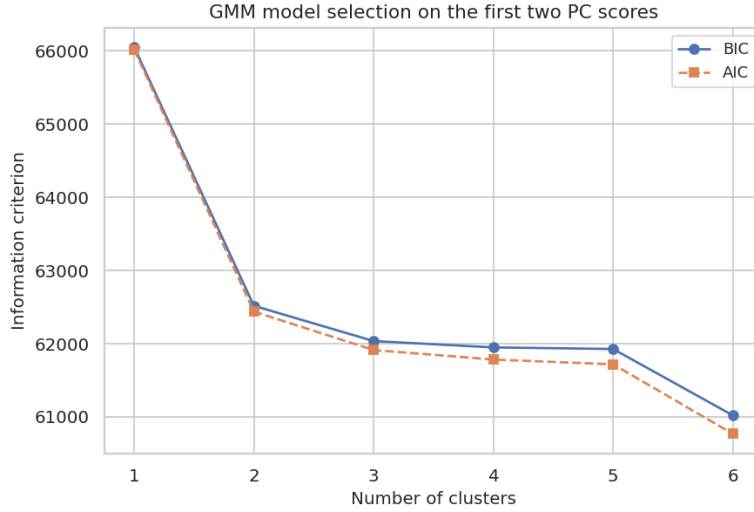


Figure A4: AIC and BIC values for Gaussian mixture models fitted to the first two principal component scores, with the number of mixture components ranging from $K = 1$ to $K = 6$. Both criteria drop sharply from one to two clusters and then decrease more gradually, reaching their smallest values at $K = 6$. This suggests that the PC score distribution contains more than a single simple cluster structure, although the gains beyond moderate values of K are relatively small.



Figure A5: Comparison of Gaussian mixture model and k -means clustering on the first two principal component scores of the standardized photometric variables. Both methods partition the data into three broad groups along the dominant left to right gradient in PCA space, indicating that the leading two components contain a strong low dimensional clustering pattern. The GMM solution produces smoother boundaries that better reflect the elliptical spread of the data, whereas k -means yields harder partitions with sharper transitions between clusters. Overall, the two methods recover a similar global structure, but the GMM fit is more flexible in accommodating unequal spread and covariance across groups.